

It can be used to attack any type of unprotected or improperly protected SQL database. This process involves entering portions of SQL code into a web form entry field – most commonly usernames and passwords – to give the hacker further access to the site backend, or to a particular user's account. When you enter logon information into sign-in fields, this information is typically converted to an SQL command. This command checks the data you've entered against the relevant table in the database. If your input data matches the data in the table, you're granted access, if not, you get the kind of error you would have seen when you put in a wrong password. An SQL injection is usually an additional command that when inserted into the web form, tries to change the content of the database to reflect a successful login. It can also be used to retrieve information such as credit card numbers or passwords from unprotected sites.

- b. **Theft of FTP Passwords:** This is another very common way to tamper with web sites. FTP password hacking takes advantage of the fact that many webmasters store their web site login information on their poorly protected PCs. The thief searches the victim's system for FTP login details, and then relays them to his own remote computer. He then logs into the web site via the remote computer and modifies the web pages as he or she pleases.
- c. **Cross-site scripting:** Also known as XSS (formerly CSS, but renamed due to confusion with cascading style sheets), is a very easy way of circumventing a security system. Cross-site scripting is a hard-to-find loophole in a web site, making it vulnerable to attack. In a typical XSS attack, the hacker infects a web page with a malicious client-side script or program. When you visit this web page, the script is automatically downloaded to your browser and executed. Typically, attackers inject HTML, JavaScript, VBScript, ActiveX or Flash into a vulnerable application to deceive you and gather confidential information.

If you want to protect your PC from malicious hackers, investing in a good firewall should be first and foremost. Hacking is done through a network, so it's very important to stay safe while using the internet. You'll read more about safety tips in the last chapter of this book.

2. Virus dissemination

Viruses are computer programs that attach themselves to or infect a system or files, and have a tendency to circulate to other computers on a network.